A Counterexample for the Dual DDD Question

Run `fa_banach_001`, lane 8

June 19, 2026

Status

This packet gives a candidate counterexample to Question 3.3 of Bellenot's *Skipped Blocking and other Decompositions in Banach spaces* [1]. The current verdict is: likely valid, pending human review.

Source Question

The source paper defines skipped blocking decompositions (SBD) and dual decomposition data (DDD). Question 3.3 appears on page 5 of the locally rendered source PDF:

So if $y \in [X_n^*]_{n>m(k+1)}$ and w is in the sphere of W, then by the construction above we can find w_i with $\|w - w_i\| < \varepsilon$ and clearly, $y(x_i) = 0$. Hence $\|w + y\| \geq \|w_i + y\| - \|w - w_i\|$ and $\|w_i + y\| \geq |(w_i + y)(x_i)|/\|x_i\| \geq 1/(1+\varepsilon)$. So we have $\|w+y\| \geq (1-\varepsilon)(1+\varepsilon)^{-1}\|w\|$ and the $R_{m(k)+1,m(k+1)}$ projection onto $[X_n^*]_{n\leq m(k)}$ with kernel $[X_n^*]_{n>m(k+1)}$ has norm bounded by $(1+\varepsilon)/(1-\varepsilon)$. □

Question 3.3. *If (X_n) is a SBD, is the dual DDD (X_n^*) already a SBD?*

Example 3.4. *A DDD (X_n) for ℓ_2, where (X_n^*) is not a SBD for $Y = [X_n^*] = \ell_2$.*

Construction. Let (e_i) be a conditional basis for ℓ_2 and let π be a permutation so that $(e_{\pi(n)})$ is not a basis. Thus the coefficient functionals $(e^*_{\pi(n)})$ are also not a basis. Letting $X_n = [e_{\pi(j)}]_{j=n}^n$, Proposition 2.7 says (X_n^*) is not a SBD. □

Corollary 3.5. *For any basis (e_n) and permutation π, the sequence $(e_{\pi(n)})$ can be blocked to be a SBD.*

The question asks whether, if (X_n) is an SBD, the dual DDD (X_n^*) is already an SBD.

Definitions Used

For a sequence (E_n) of finite-dimensional subspaces, write

$$E[a,b] = [E_n]_{n=a}^b.$$

The relevant SBD test is the uniform boundedness of the skipped projections

$$R_{m,k} : E[1,m-1] \oplus E[k+1,\infty) \longrightarrow E[1,m-1], \qquad m \leq k.$$

In Hilbert-space examples below, if the two summands in the domain are orthogonal then $\|R_{m,k}\| = 1$.

For a DDD (X_n) in a Banach space X, Bellenot defines

$$X_n^* = [X_j]_{j \neq n}^{\perp} \subset X^*.$$

In a Hilbert space we identify X^* with X.

Proof Intuition

The source already shows that a general DDD can have a dual DDD which is not an SBD, but in that example the original sequence is not an SBD. The construction here inserts a very small middle perturbation into orthogonal three-dimensional Hilbert blocks. In the original space, every skipped cut sees only orthogonal before and after pieces, so the SBD constant is 1. Biorthogonality reverses the small middle perturbation: two dual lines become almost parallel after the middle dual line is skipped. The corresponding skipped projection constants therefore tend to infinity.

Theorem 1. *There is a skipped blocking decomposition (X_n) of a separable Hilbert space such that the dual* DDD (X_n^*) *is not an* SBD*.*

Proof. Let (ε_r) be a positive sequence with $\varepsilon_r \downarrow 0$, for instance $\varepsilon_r = 2^{-r}$. Let

$$H = \left(\bigoplus_{r=1}^{\infty} H_r \right)_2, \qquad H_r = \mathbb{R}^3,$$

and let $e_{r,1}, e_{r,2}, e_{r,3}$ be the standard orthonormal basis of H_r. Define one-dimensional subspaces in the order

$$X_{3r-2} = [e_{r,1}], \qquad X_{3r-1} = [e_{r,1} + \varepsilon_r e_{r,2} + e_{r,3}], \qquad X_{3r} = [e_{r,3}].$$

Within each H_r, the three displayed vectors are linearly independent; across different r's the blocks are orthogonal. Hence (X_n) is total and minimal, so it is a DDD for H.

We now check that (X_n) is an SBD. Fix a skipped interval $[m, k]$. The space $X[1, m-1]$ and the tail $X[k+1, \infty)$ are orthogonal. Indeed, different H_r's are orthogonal. Inside a single three-dimensional block, the only way for both a before-piece and an after-piece to remain after deleting a consecutive skipped interval is to delete the middle line X_{3r-1}, leaving $X_{3r-2} = [e_{r,1}]$ before and $X_{3r} = [e_{r,3}]$ after; these two lines are orthogonal. Thus every skipped projection $R_{m,k}$ is an orthogonal projection, and

$$\sup_{m \leq k} \|R_{m,k}\| = 1.$$

Therefore (X_n) is an SBD.

It remains to compute the dual DDD. In the Hilbert identification $H^* = H$, the annihilator lines in block r are

$$Y_{3r-2} = X_{3r-2}^* = [u_r], \quad u_r = e_{r,1} - \varepsilon_r^{-1} e_{r,2},$$

$$Y_{3r-1} = X_{3r-1}^* = [e_{r,2}],$$

and

$$Y_{3r} = X_{3r}^* = [z_r], \quad z_r = e_{r,3} - \varepsilon_r^{-1} e_{r,2}.$$

For example, u_r annihilates X_{3r-1} and X_{3r} but not X_{3r-2}, and similarly for the other two lines. These dual lines span each H_r, so their closed span is again H.

Now consider in the dual DDD the skipped interval consisting only of the middle line Y_{3r-1}. The associated skipped projection maps

$$Y[1, 3r-2] \oplus Y[3r, \infty) \longrightarrow Y[1, 3r-2].$$

Restricted to the two-dimensional subspace $[u_r] \oplus [z_r]$, it sends $u_r - z_r$ to u_r. Hence its norm is at least

$$\frac{\|u_r\|}{\|u_r - z_r\|} = \frac{\sqrt{1+\varepsilon_r^{-2}}}{\sqrt{2}}.$$

This lower bound tends to infinity as $r \to \infty$. Therefore the skipped projection constants of the dual DDD are unbounded, so (X_n^*) is not an SBD. □

Verification Notes

The script `code/check_three_block_constants.py` checks the local three-dimensional constants for $\varepsilon = 2^{-k}$, $1 \le k \le 8$. It confirms that the primal middle-skip projection has norm 1, while the dual middle-skip projection grows like $1/\varepsilon$. The command used was:

```
conda run --no-capture-output -n sandbox python code/check_three_block_constants.py.
```

This numerical check is not used as proof; it is only a guard against mistakes in the finite-dimensional linear algebra.

Novelty and Literature Check

The cheap run indexes were searched for `0408004`, `skipped blocking`, `SBD`, `dual DDD`, and `predecomposition space`; no exact prior packet was found. Web searches on June 19, 2026 used the exact phrases `"If (X_n) is a SBD" "dual DDD"`, `"Skipped Blocking" "dual DDD" "SBD"`, and `"predecomposition space" "SBD"`. They found the source paper and unrelated later uses of skipped blocking, but no later answer to Question 3.3.

The source's Example 3.4 is not a counterexample to Question 3.3, because there the original sequence is only a DDD, not an SBD. The present construction keeps the original sequence an SBD with constant 1.

Human Review Recommendation

Review should focus on three points: the global claim that all primal skipped before/after pieces are orthogonal; the identification of the three dual annihilator lines in each Hilbert block; and the lower bound for the dual middle-skip projection constants.

References

[1] S. F. Bellenot, *Skipped Blocking and other Decompositions in Banach spaces*, arXiv:math/0408004, 2004.